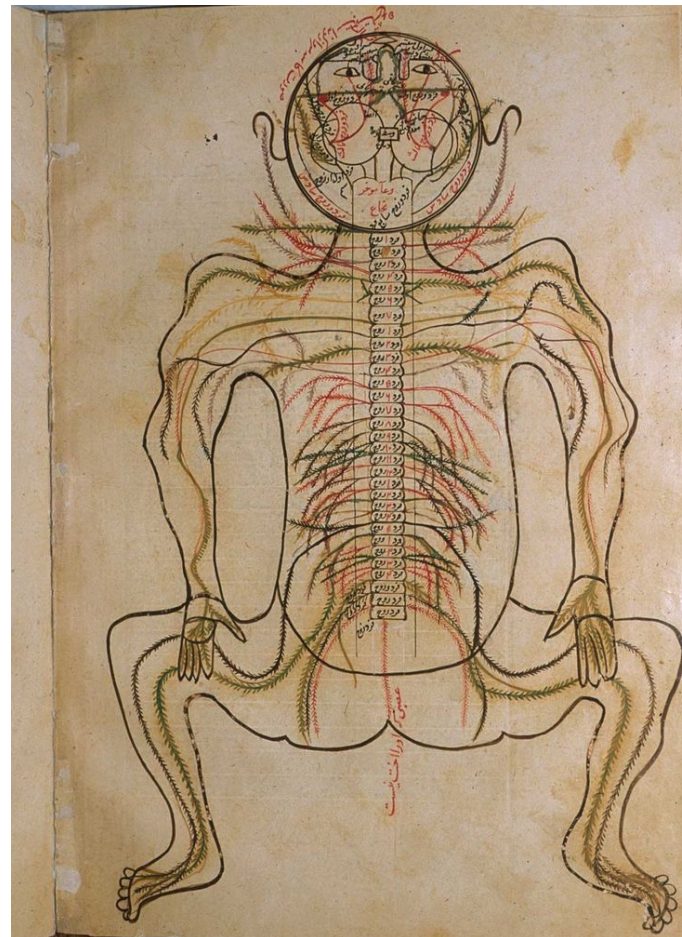


Peripheral Nervous System (PNS) and Autonomic nervous system (ANS)

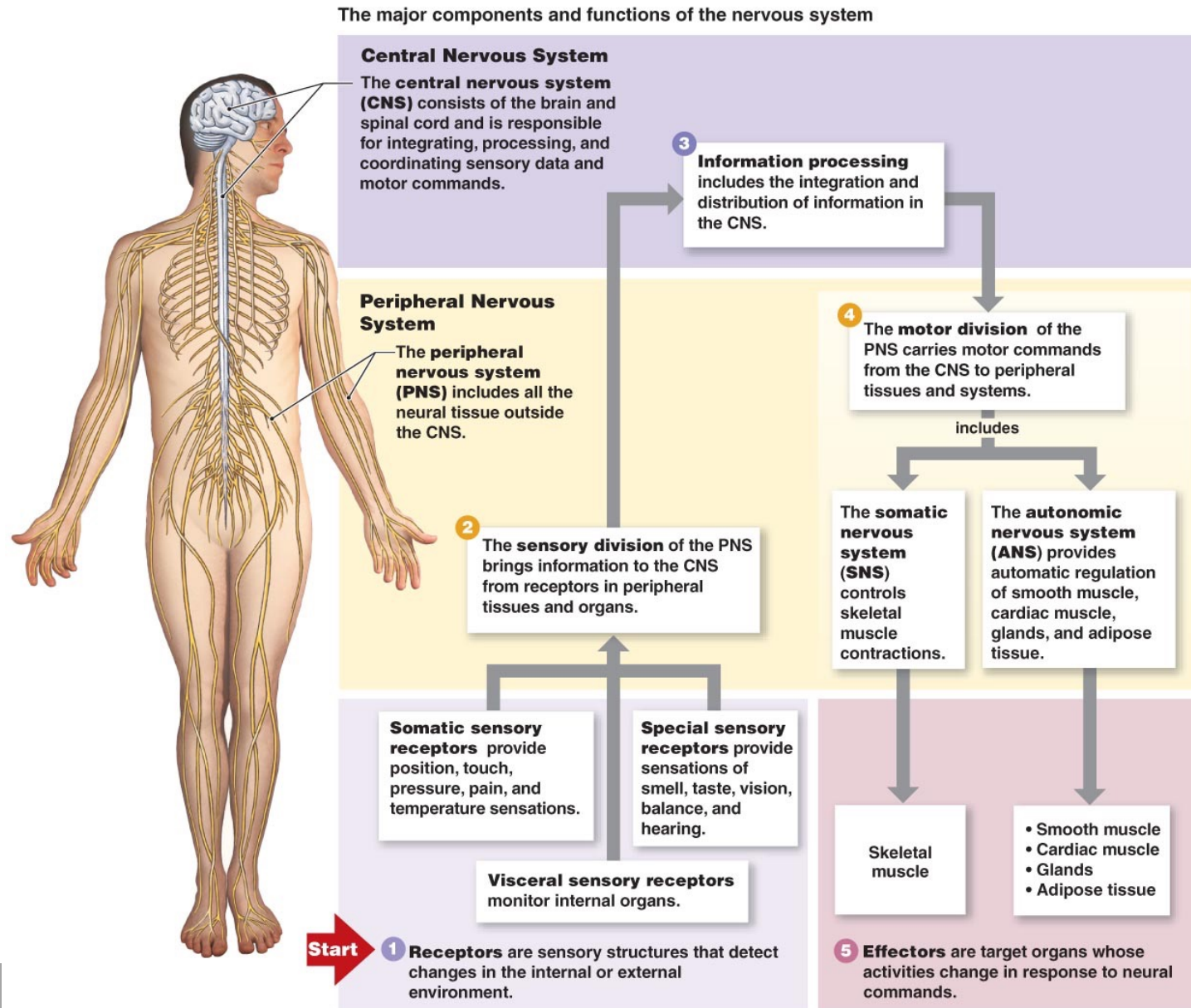


*Anatomical illustration of the nervous system
by Mansour Ibn-Ilyas, ca. 1400*

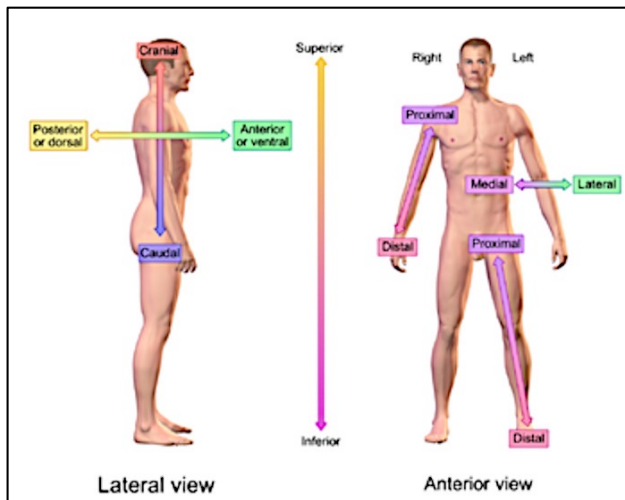
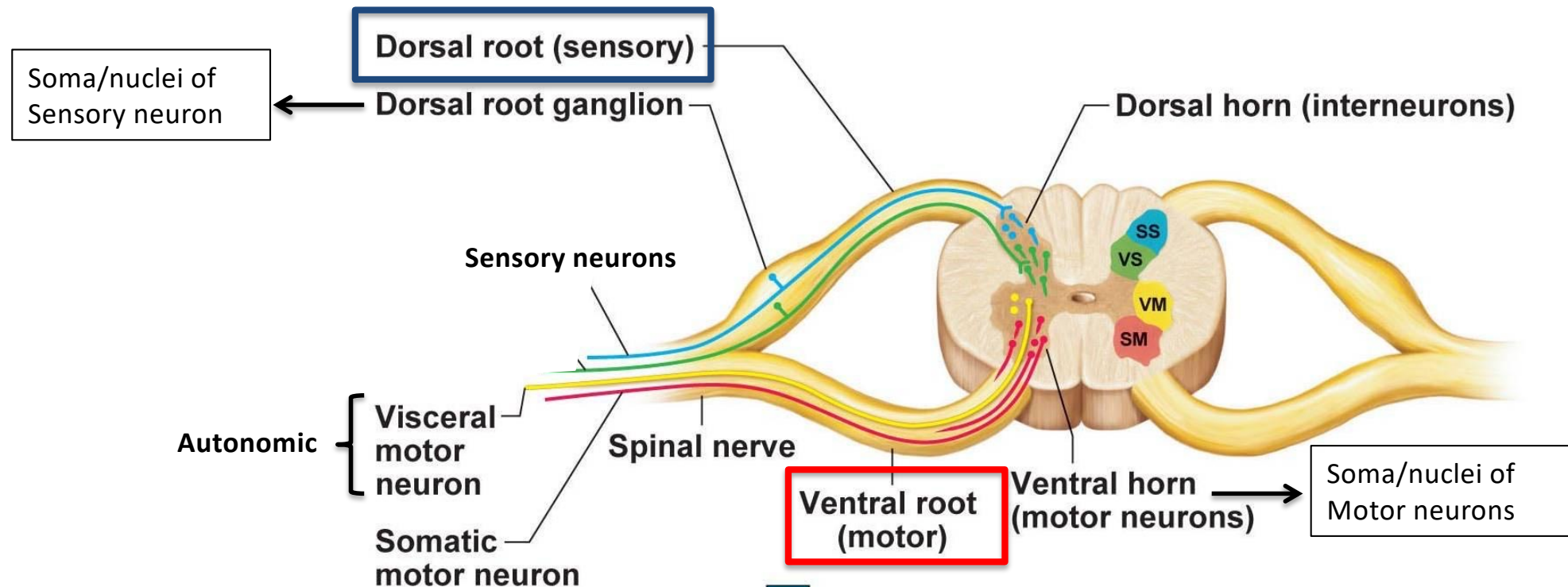
Agenda

1. Organization of the Peripheral nervous System (PNS)
2. Neuronal Fibers, Nerves and Plexuses
3. Autonomic nervous system:
 - Sympathetic nervous system
 - Parasympathetic nervous system
 - Adrenal medulla

Sensory and Motor/Effector Divisions of the PNS



Nerves centrally divide in sensory and “motor” roots

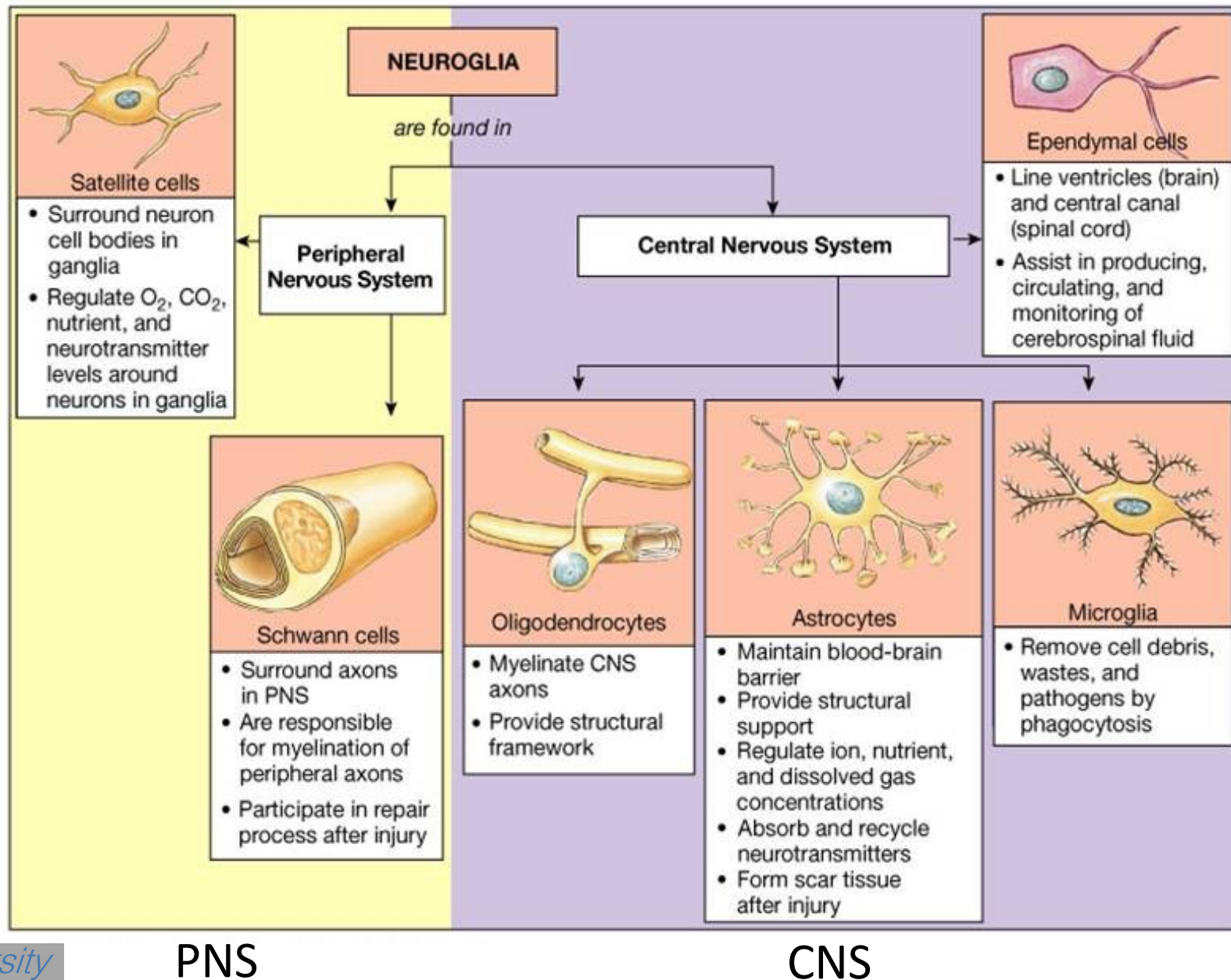


- SS** Interneurons receiving input from somatic sensory neurons
- VS** Interneurons receiving input from visceral sensory neurons
- VM** Visceral motor (autonomic) neurons
- SM** Somatic motor neurons

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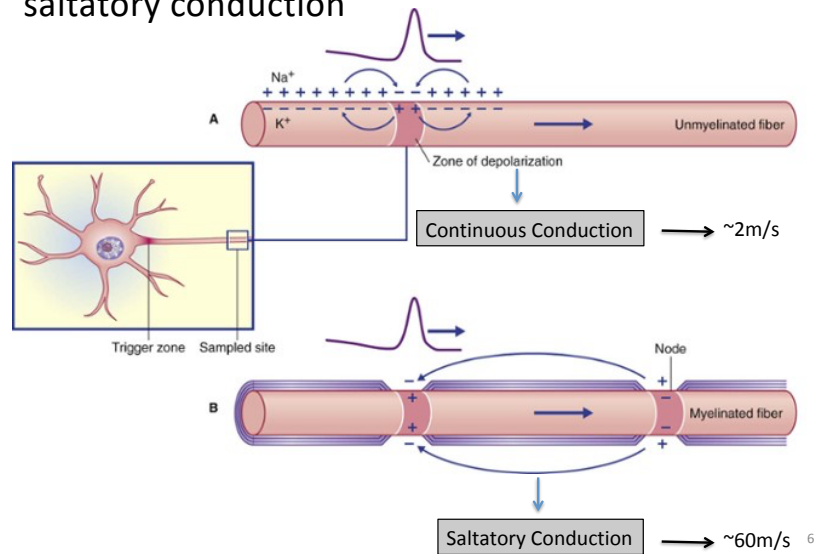


Not only neurons: Neuroglia in PNS

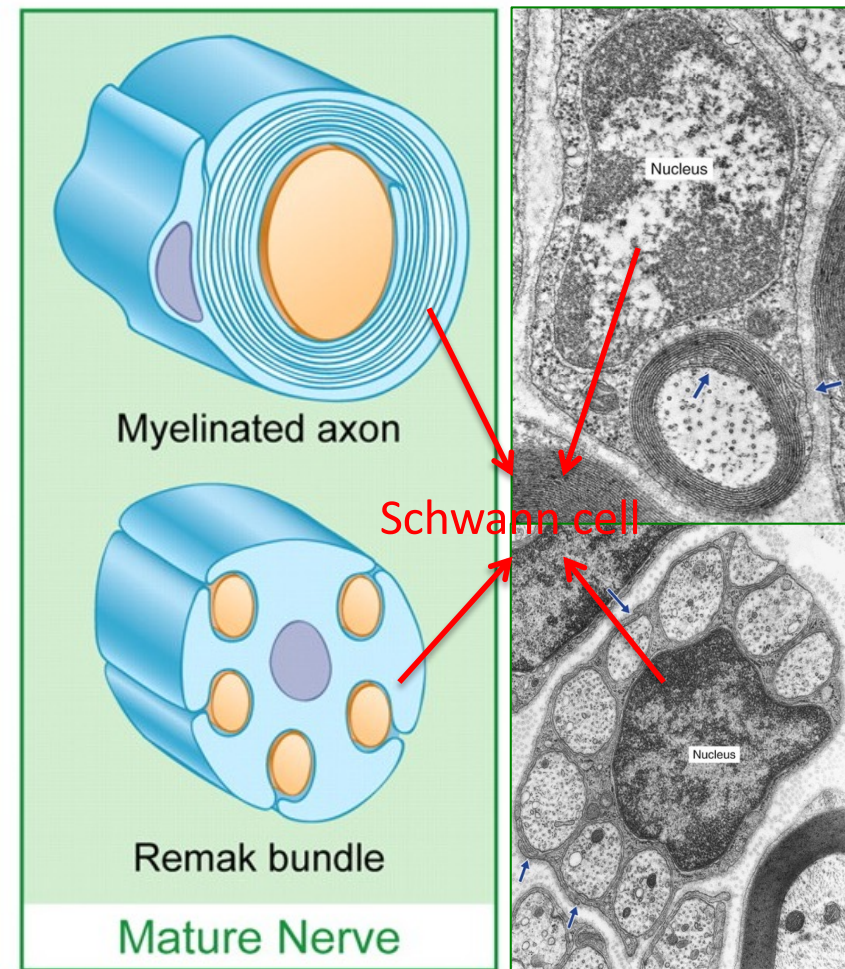


Myelinated and unmyelinated fibers

- **Myelin consists in multiple insulating layers** that facilitate saltatory conduction



- **All Axons are associated with Schwann cells, but less than half** are myelinated
- **Remak fibers are unmyelinated.** Still associated with Schwann cells, but :
 - Only partially covered
 - Max two turns of plasma membrane
 - Reduced regeneration
 - Reduced speed of conduction
- **In humans there are ~ 1-4 Remak fibers per Schwann unit (Remak bundle).** In small animals much more.

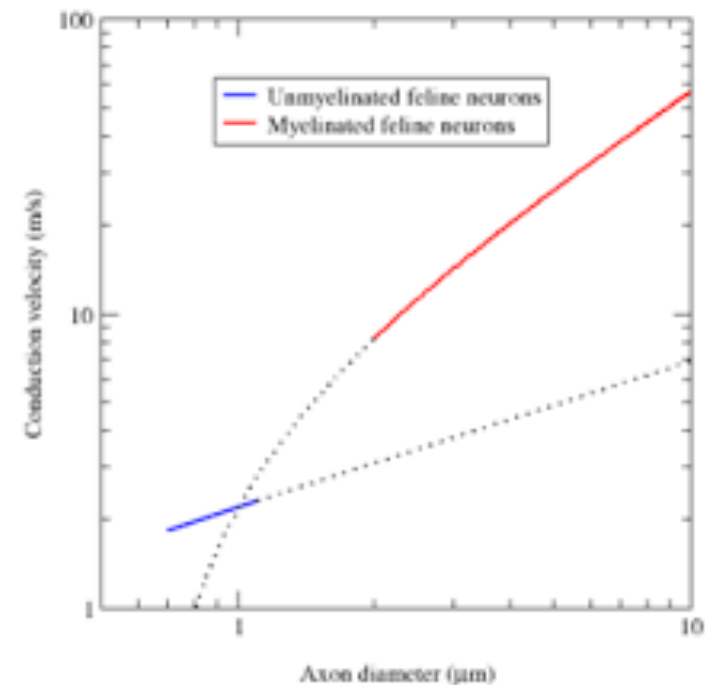
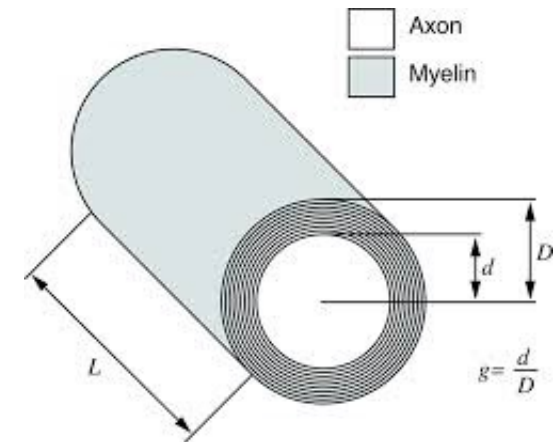


Dimensions make the difference in fibers

Myelin thickness is dependent on axon diameter



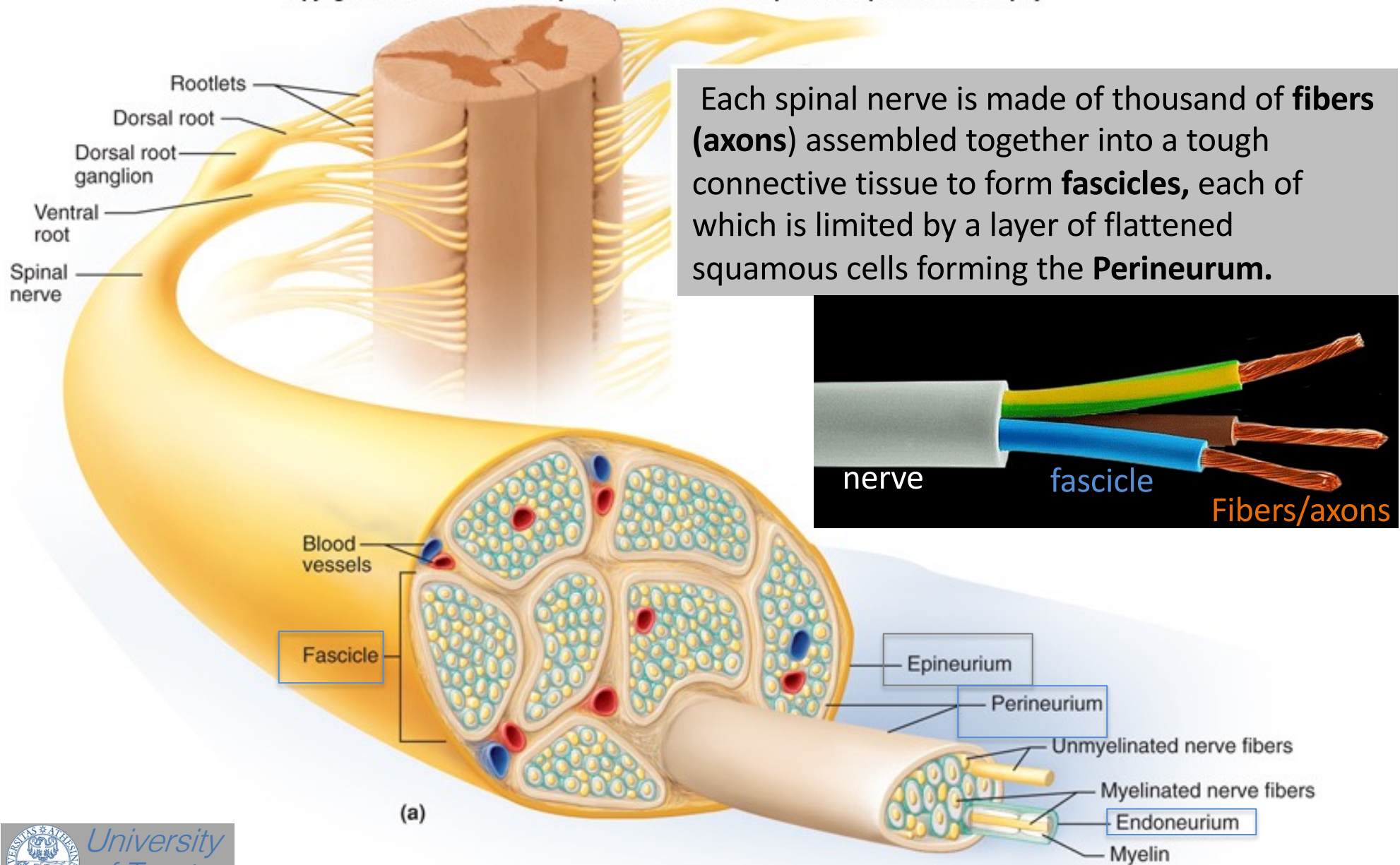
**There is a strong correlation between the diameter of axon and speed of conduction
→ classification of Fiber types**





Axons are organized in Fascicles inside the nerves

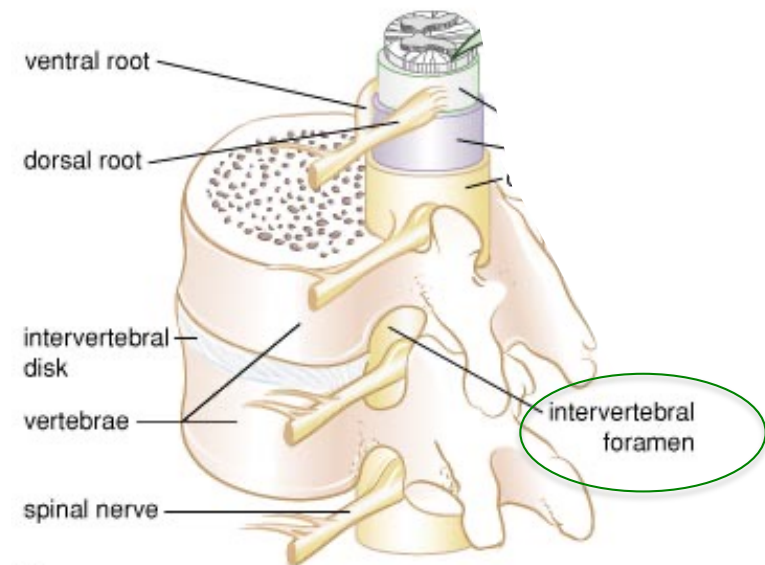
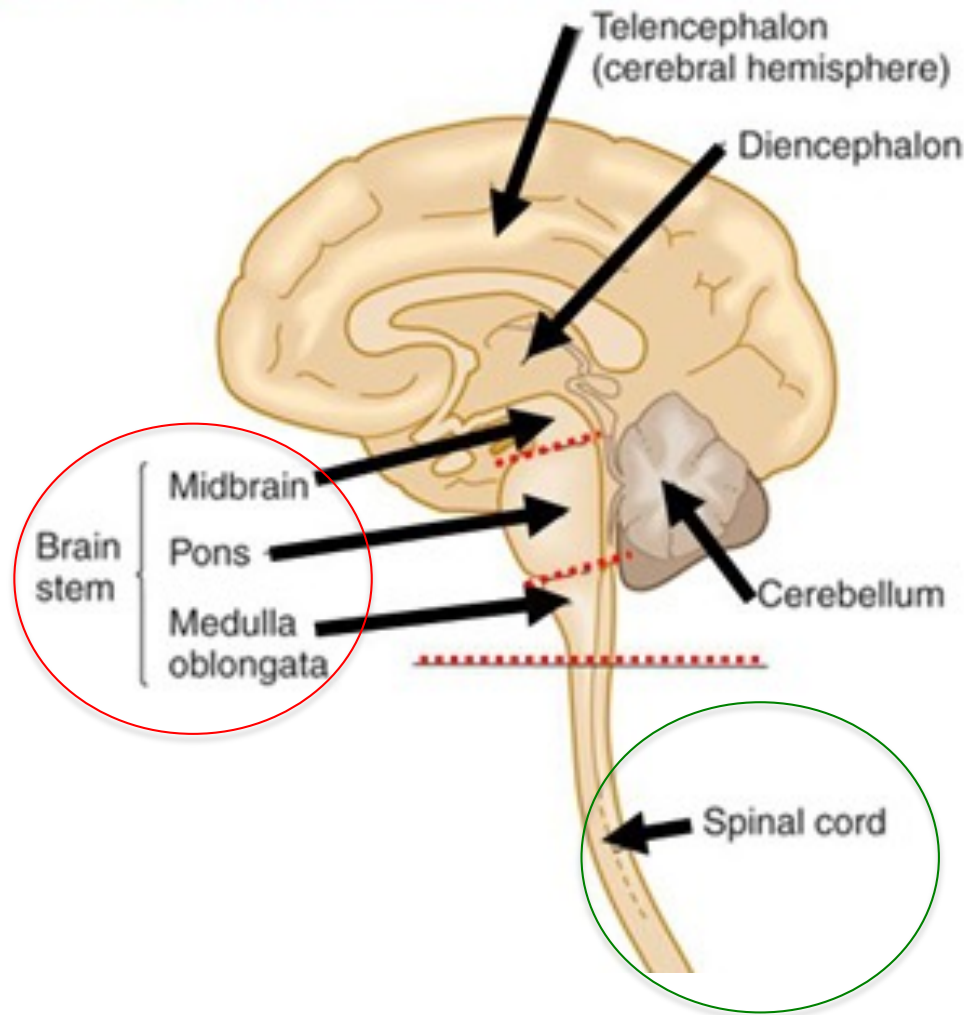
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Peripheral Nerves

- **10 Pairs of CRANIAL NERVES** → Brain Stem
+2 (optic and olfactory) that are fully part of CNS

- **31 Pairs of SPINAL NERVES** → Spinal cord



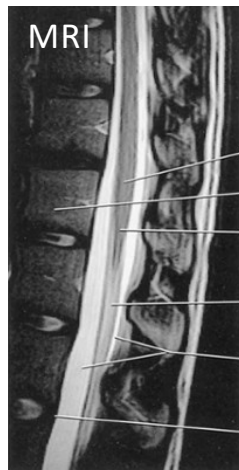
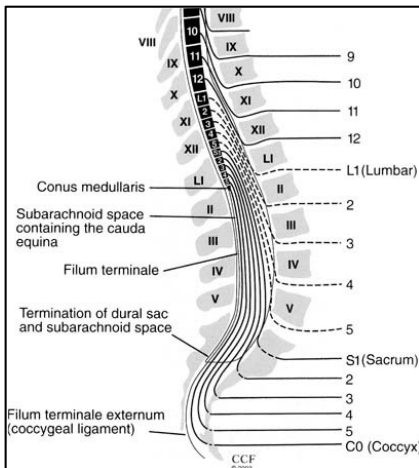
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Spinal Nerves

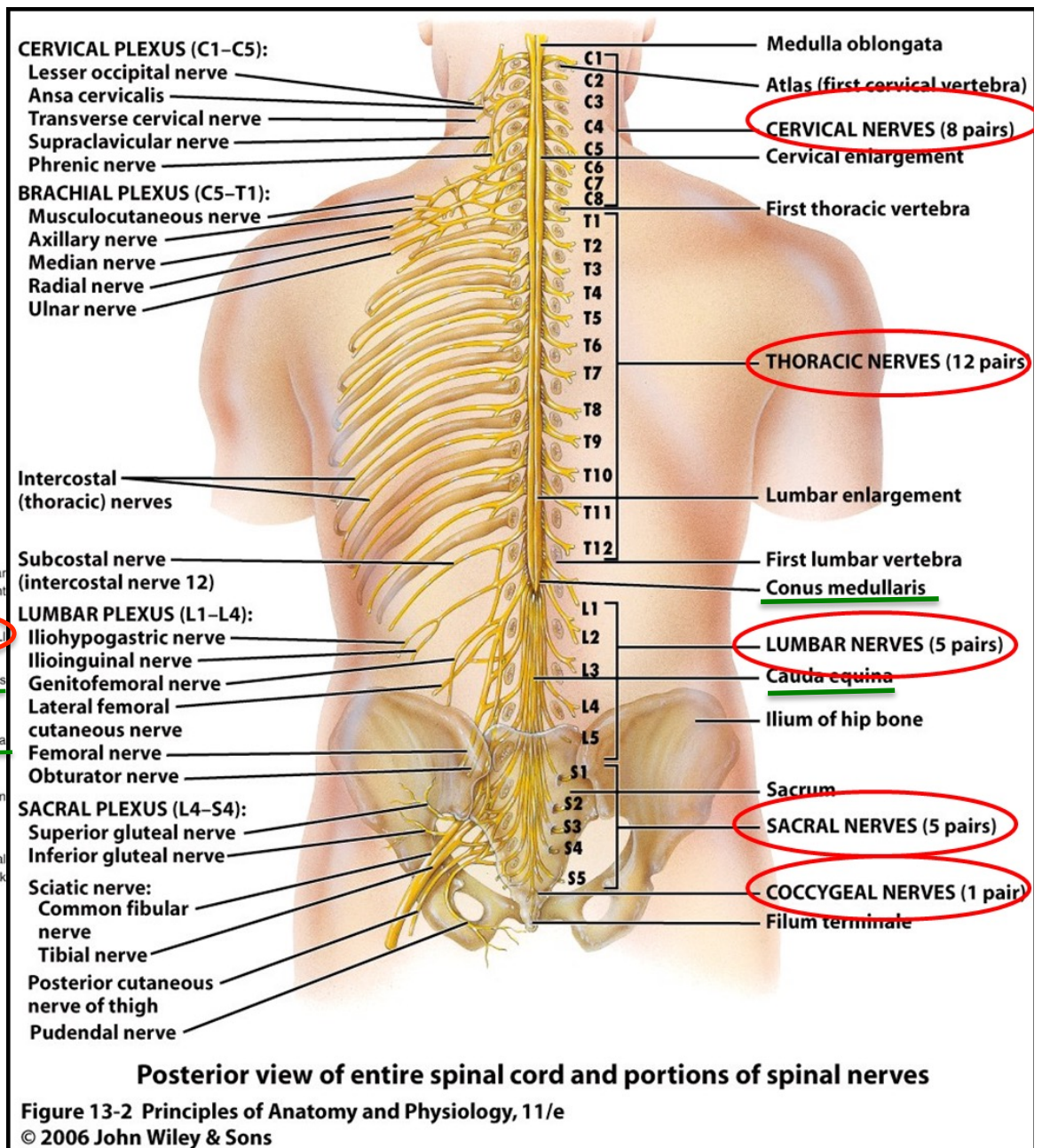
Spinal nerves are named after the section of the spine they come out of:

- 8 pairs of cervical nerves (C1-C8)
- 12 pairs of thoracic nerves (T1 – T12)
- 5 pairs of lumbar nerves (L1-L5)
- 5 pairs of sacral nerves (S1-S5)
- 1 pair of coccygeal nerves (Co1)

Conus medullaris and cauda equina



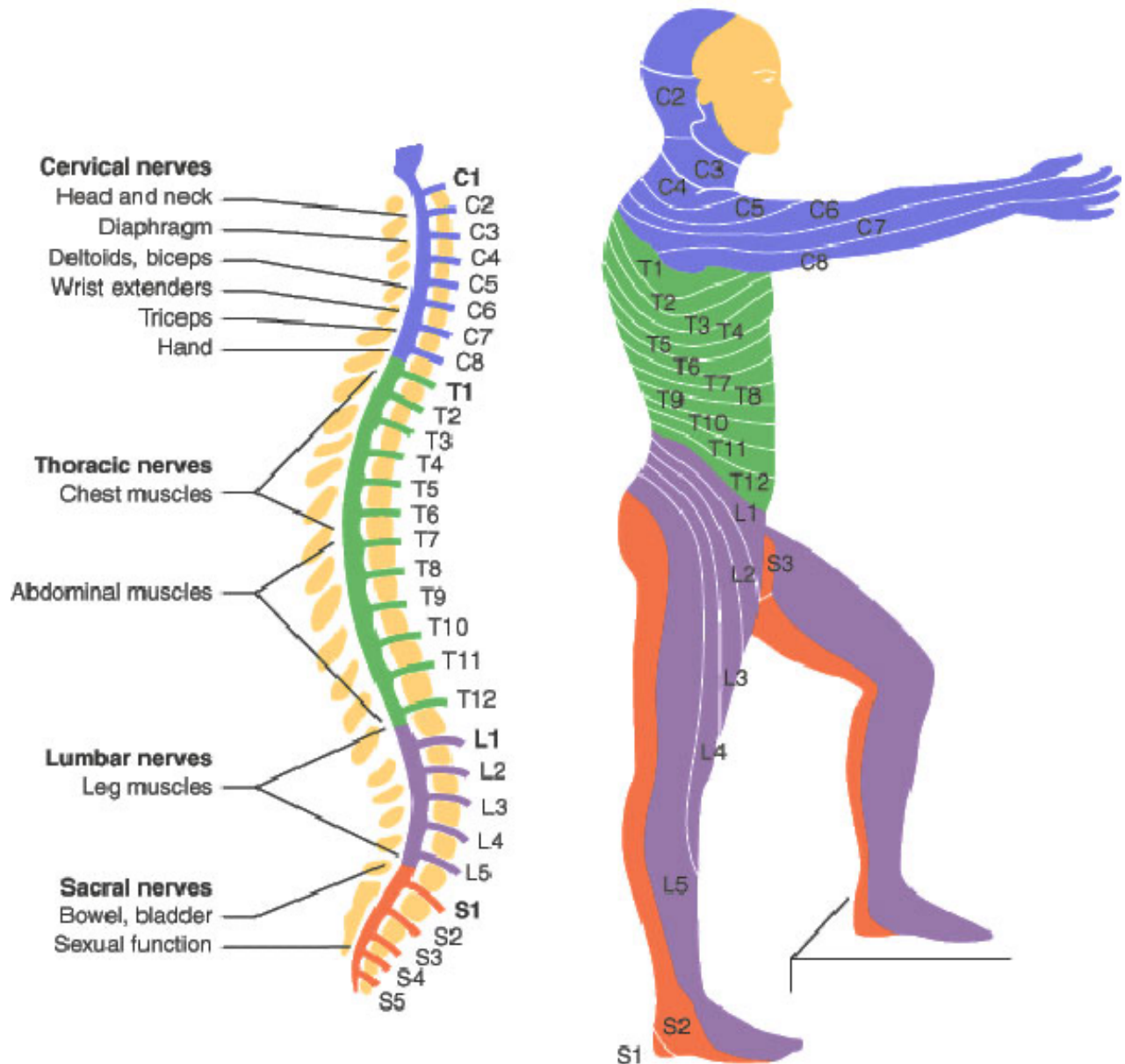
Spinal cord (**conus medullaris**) terminates at ~ level L1/L2 → lumbar puncture to collect **cerebrospinal fluid** is performed between two vertebrae at level L3/L4, or L4/L5, where there is no risk of accidental injury



Segmental distribution of Spinal Innervation

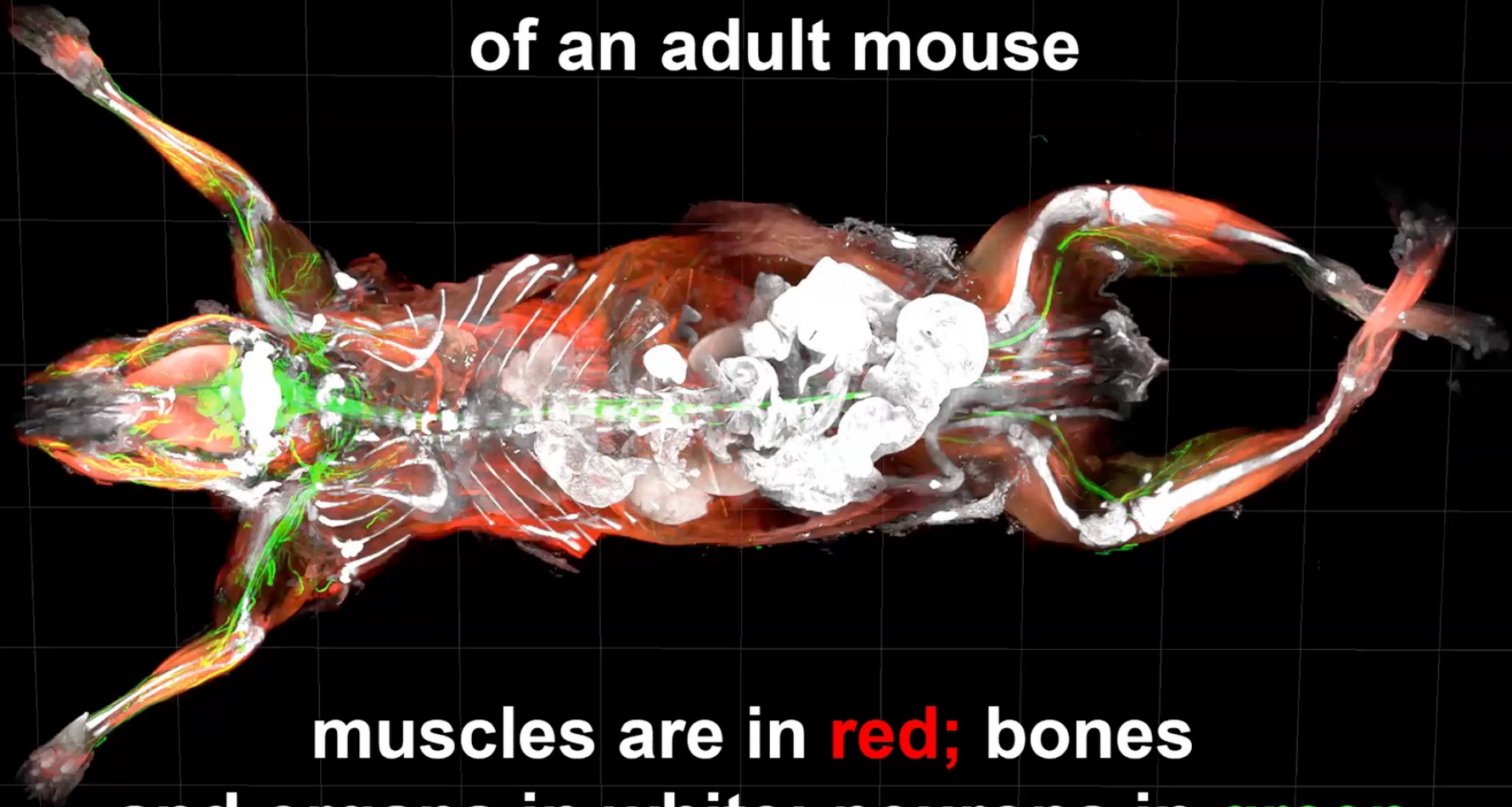
Dermatome is the **strip of skin** supplied by an **individual spinal nerve**

Borders are not sharp:
successive nerves partially overlap in the same segments



Whole body nerve distribution in mouse

**whole-body neuronal connectivity map
of an adult mouse**



**muscles are in red; bones
and organs in white; neurons in green**

Nerve Plexuses

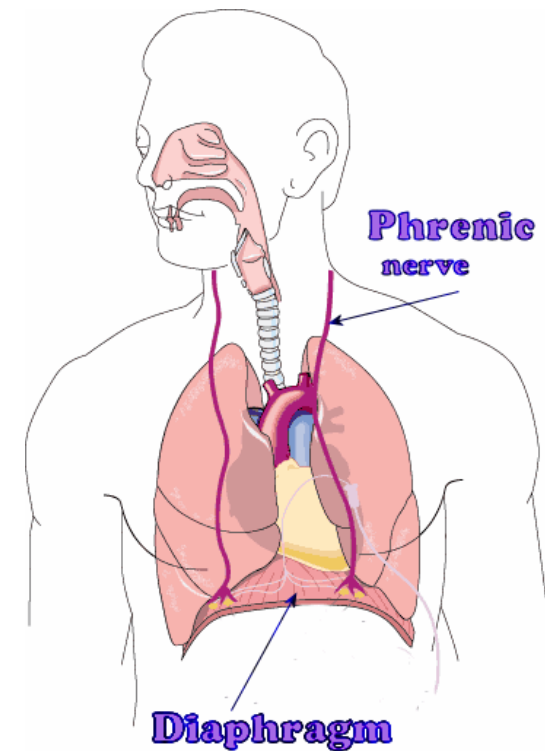
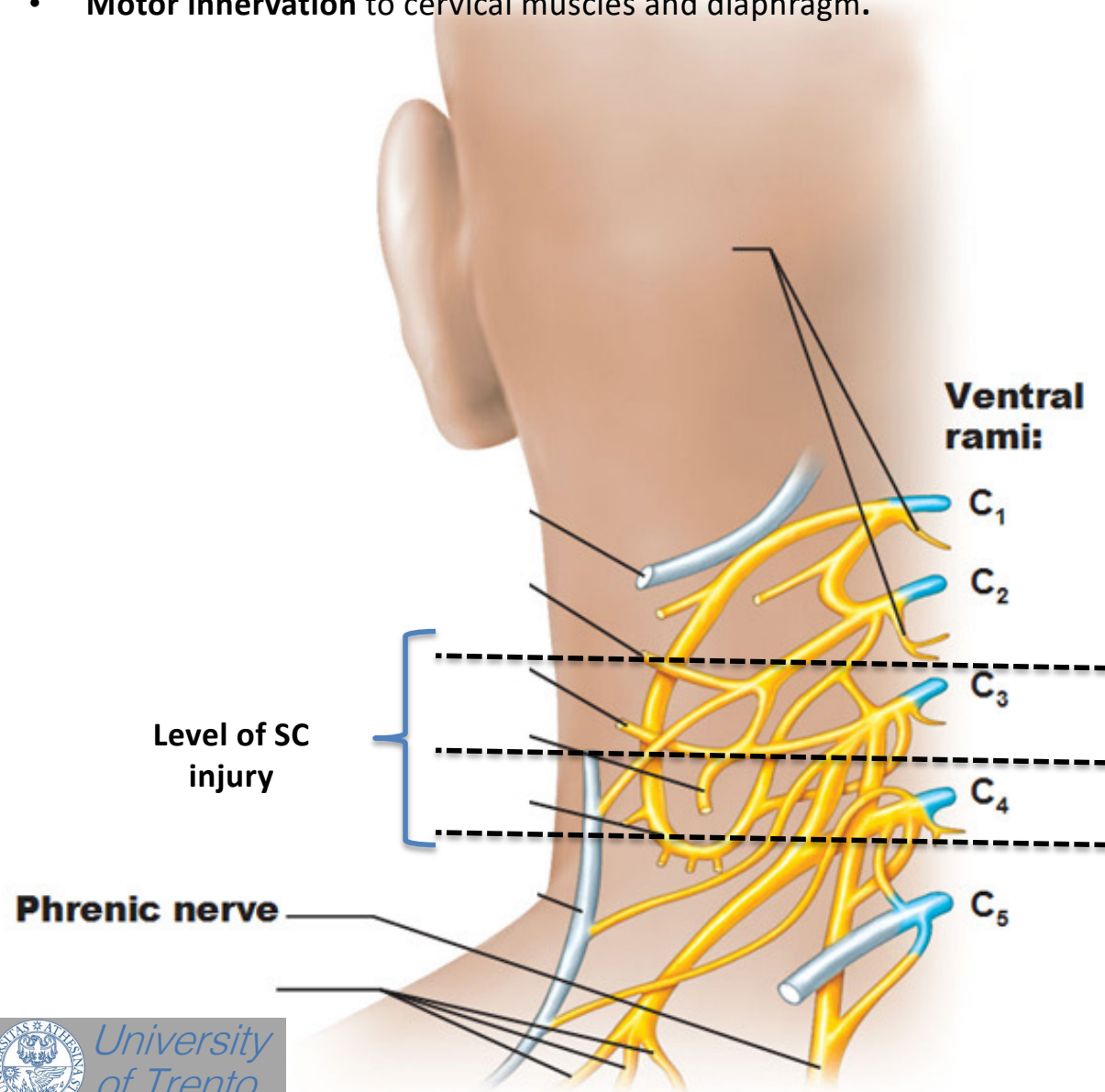
- **A nerve plexus is a network of nerves** that come together and then redistribute themselves out with a different distribution of nerves.
- **Each end of each plexus (terminal nerve) contains fibers from several spinal nerves.**
- **Four plexuses: Cervical, Brachial, Lumbar and Sacral**



Saigon

Cervical Plexus (C1→C5)

- **Sensory** from scalp, neck, and chest, muscle tendons.
- **Motor innervation** to cervical muscles and diaphragm.



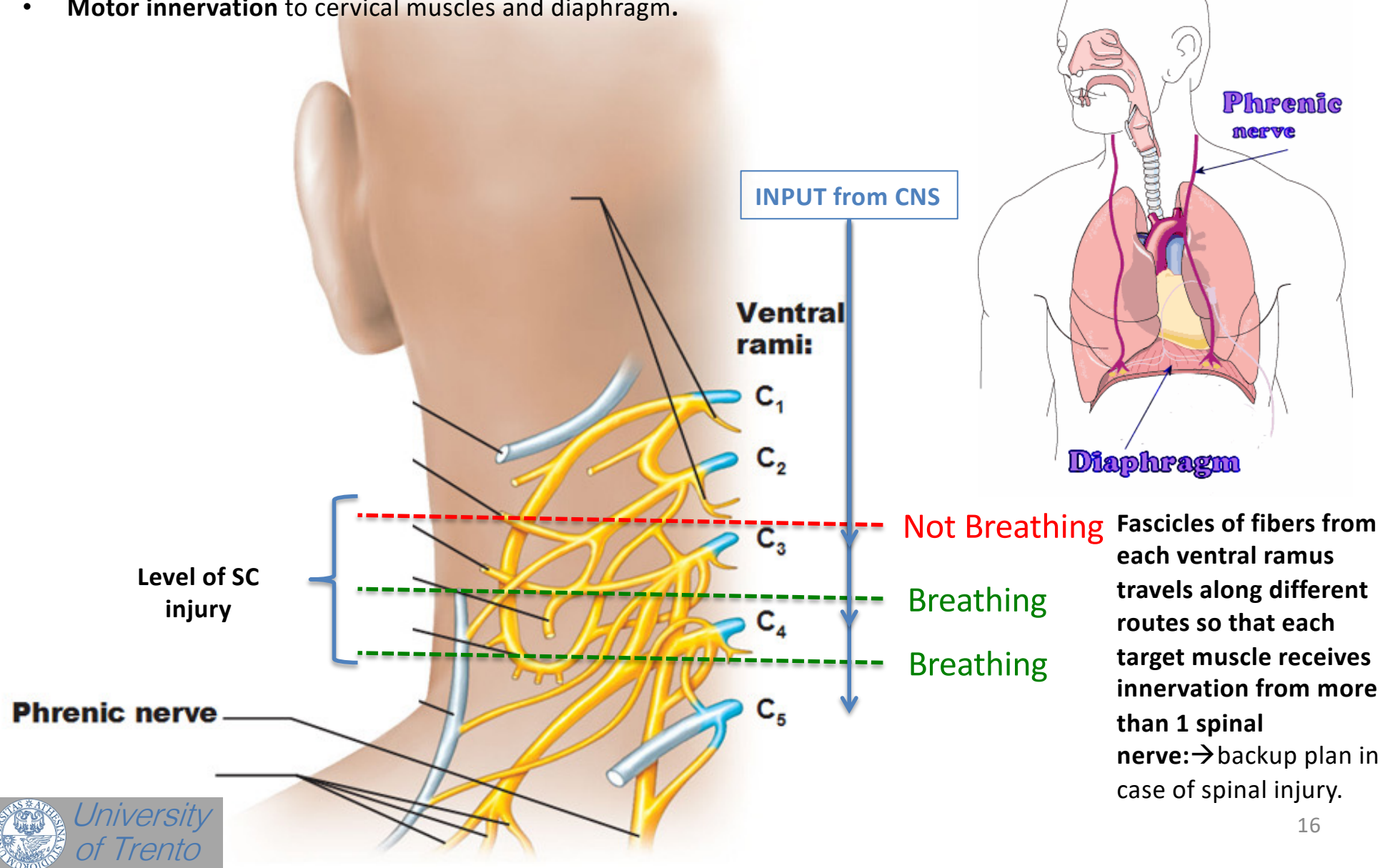
Is the
person still
breathing

?

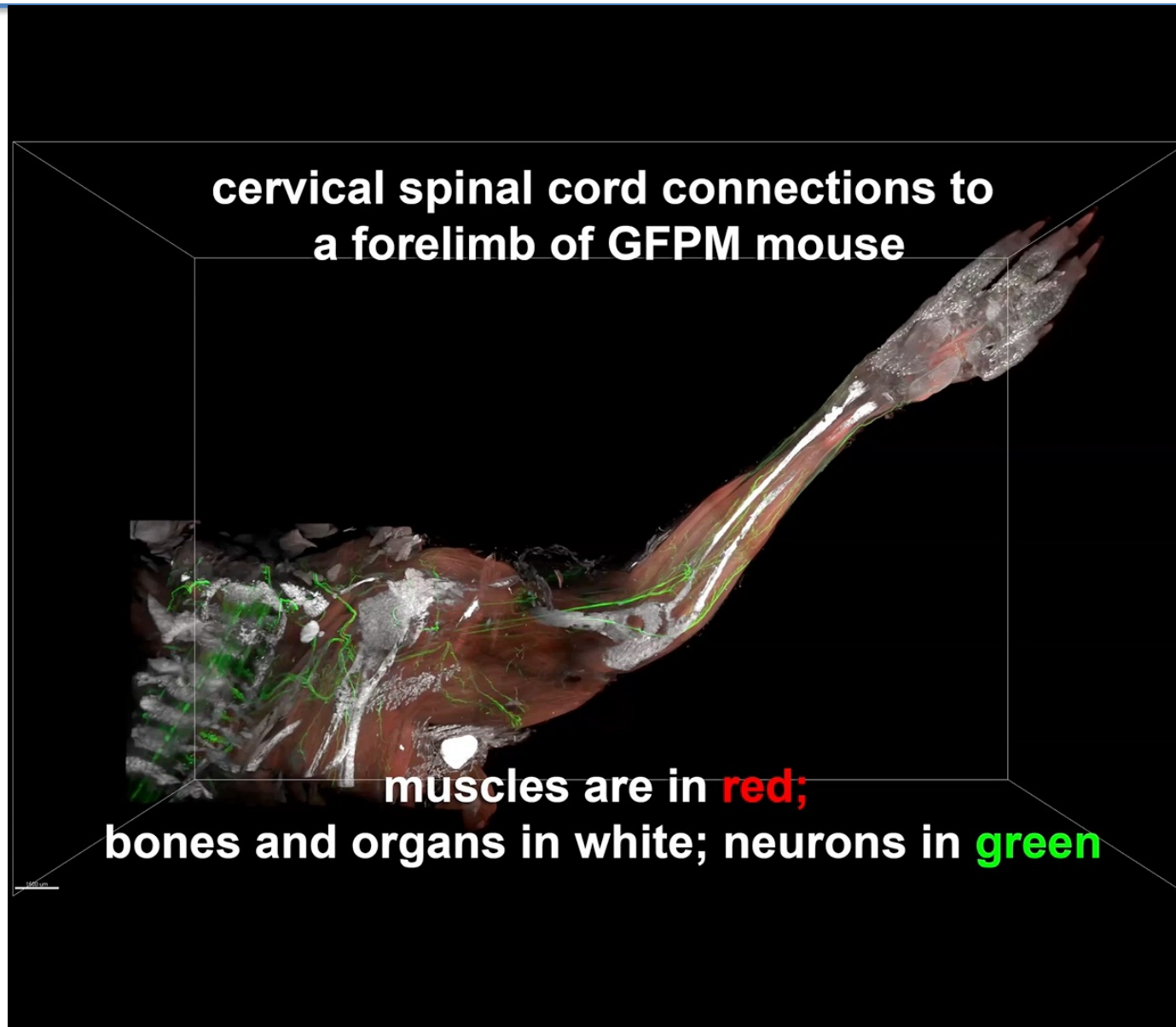
- The **CNS is controlling the phrenic nerve** by projecting through the SC to C3, C4, C5
- Efferent fibers from **C3, C4, and C5 converge** in the phrenic nerve

Cervical Plexus (C1-C5)

- **Sensory** from scalp, neck, and chest, muscle tendons.
- **Motor innervation** to cervical muscles and diaphragm.



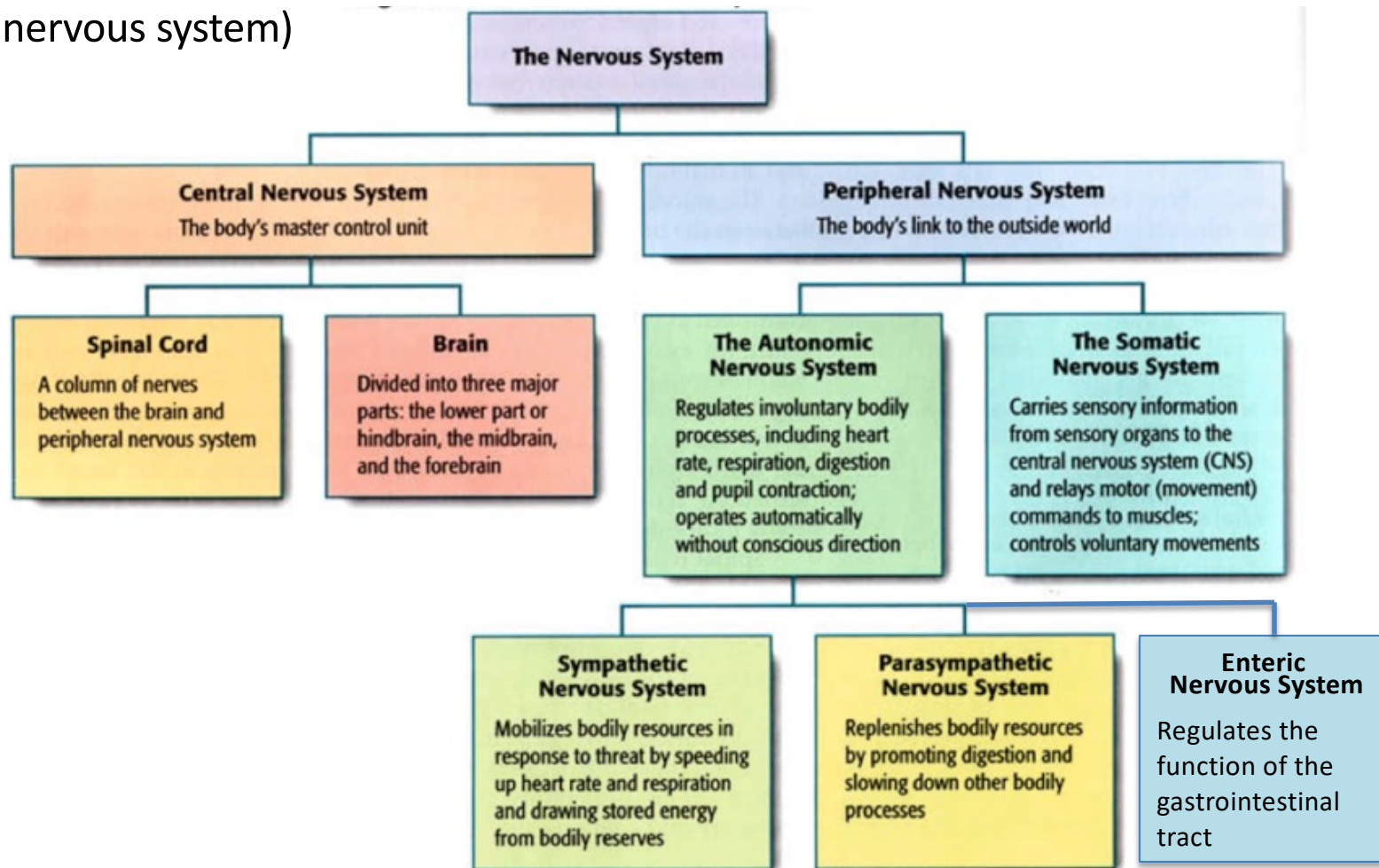
Brachial Plexus in mouse



Autonomic nervous system

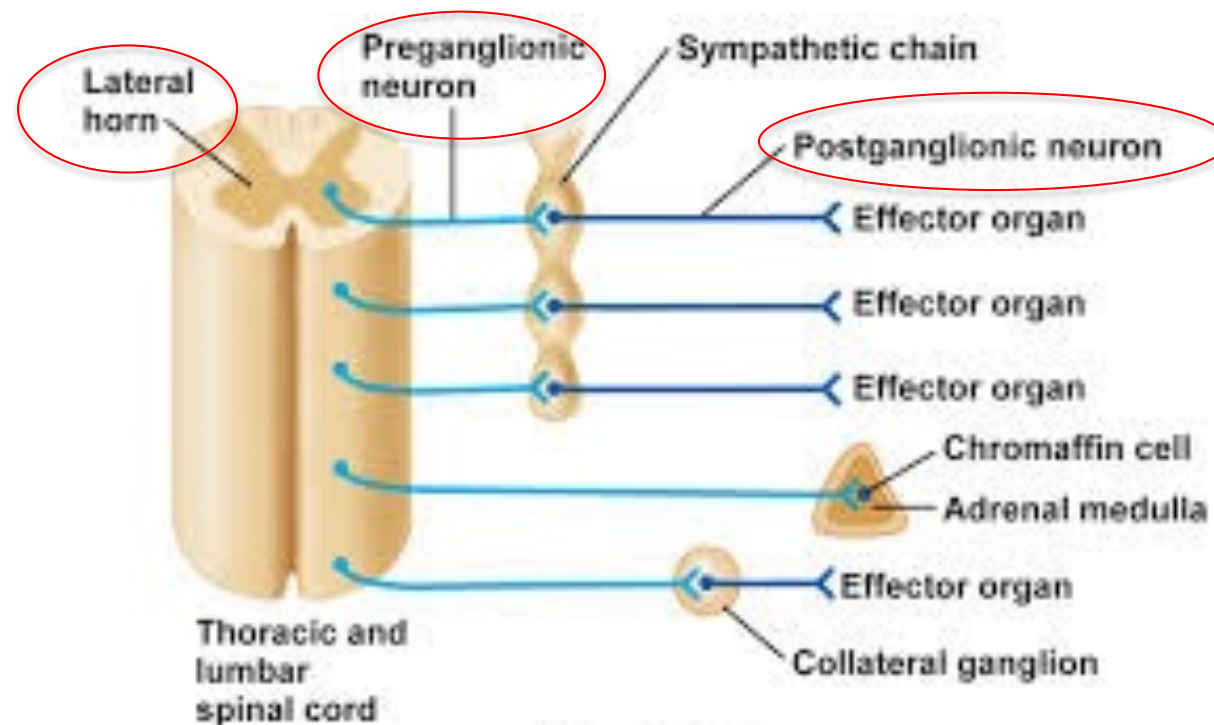
The autonomic nervous system is anatomically and functionally divided in major branches :

- Sympathetic
- Parasympathetic
- (Enteric nervous system)

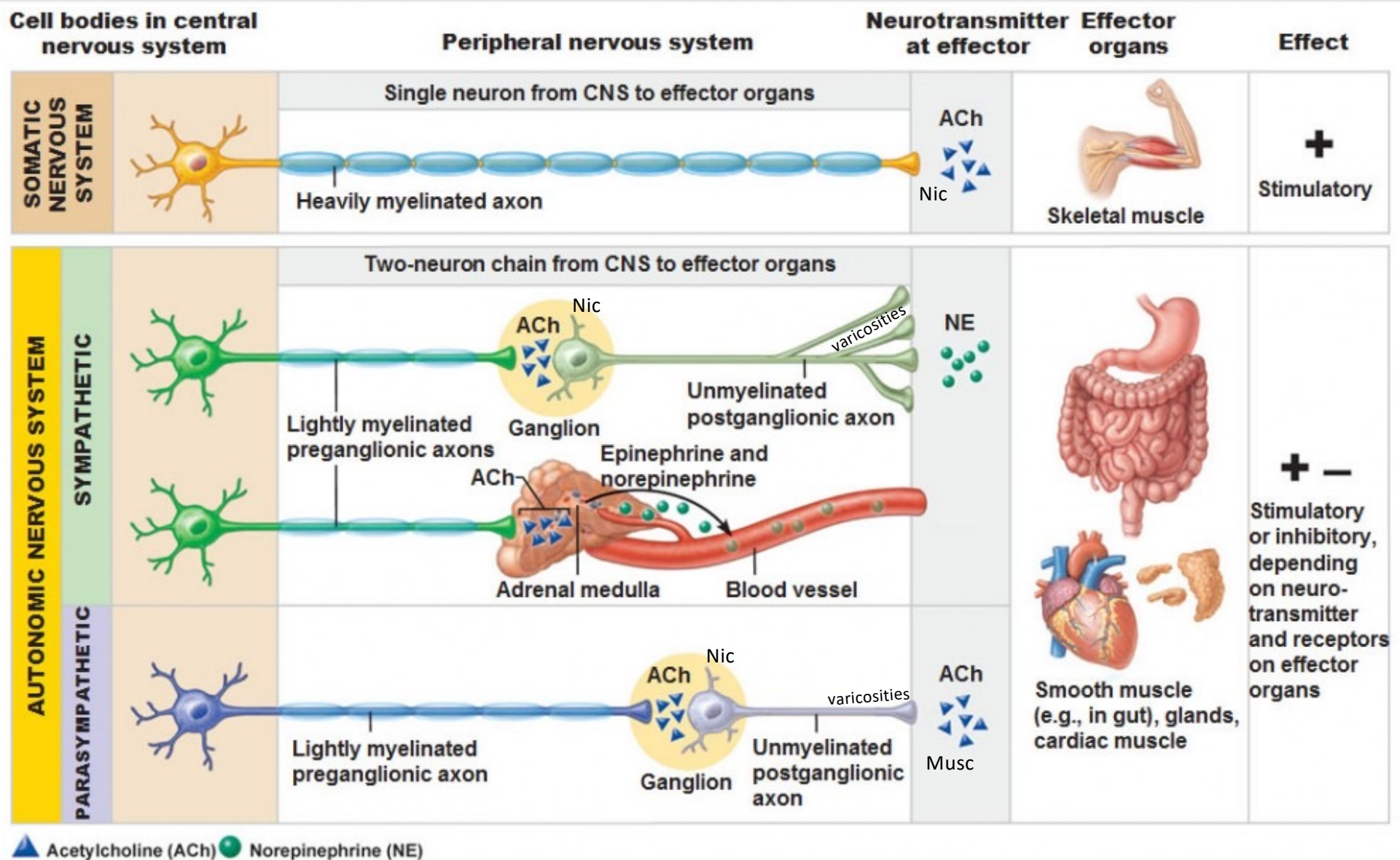


Autonomic Nervous System and Nerves

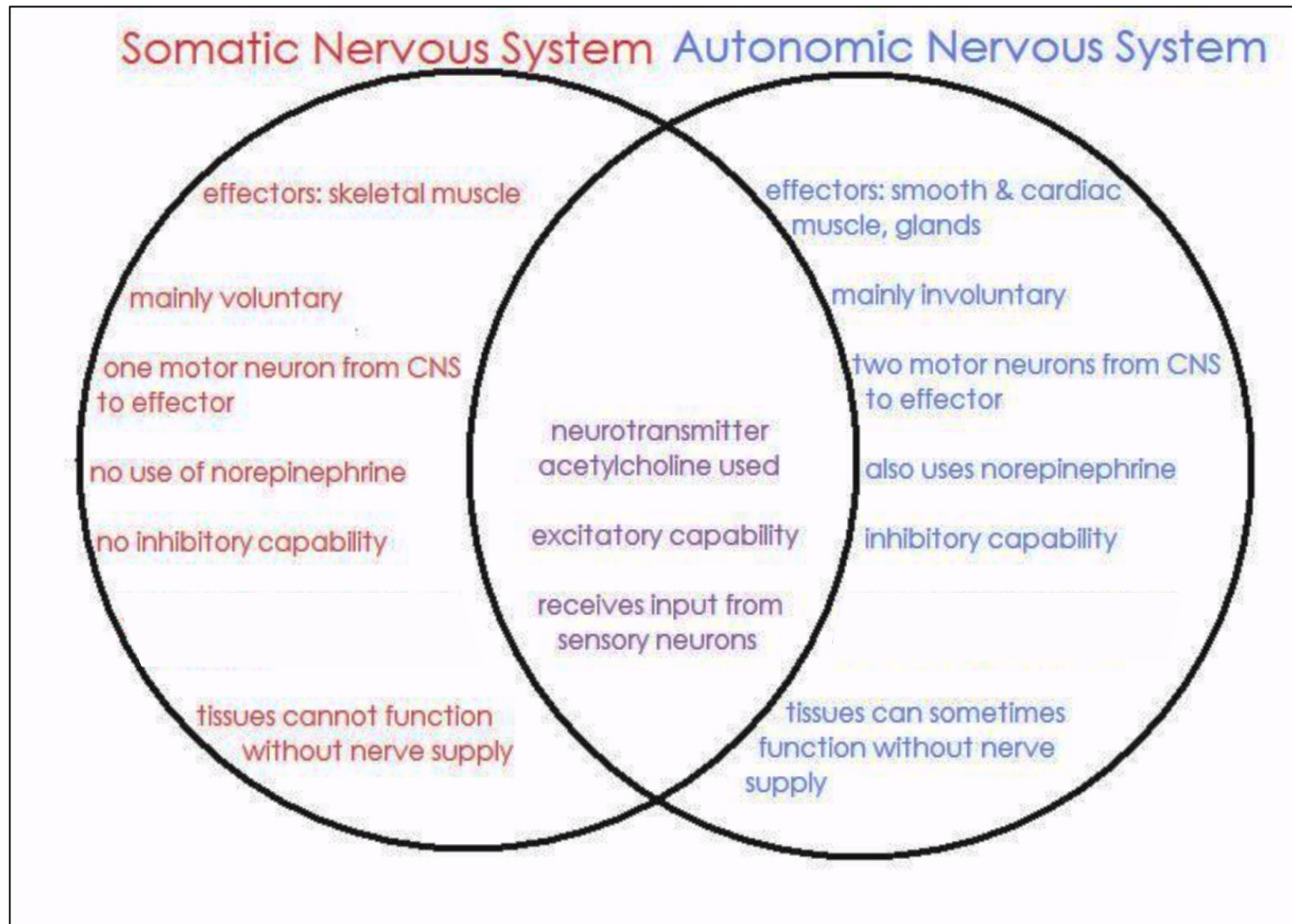
- The **autonomic** ('self-regulating' or involuntary) nervous system is distributed to the peripheral tissues and organs by way of outlying autonomic ganglia.
- **Controlling centers mainly in the hypothalamus and brainstem send central autonomic fibers to synapse** upon preganglionic neurons located in **the gray matter of the brainstem and spinal cord (lateral horn)**.
- From these neurons, **preganglionic fibers** project out of the CNS to synapse upon neurons in the autonomic ganglia. **Postganglionic fibers** emerge and form terminal networks in the target tissues.



Comparison of Autonomic and Somatic Motor System

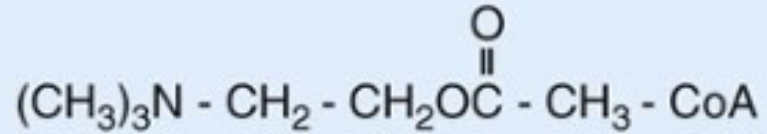


Somatic and Autonomic NS: differences and similarities



Major ANS neurotransmitters

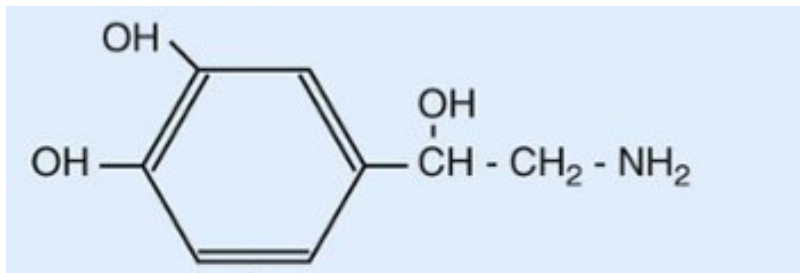
- **Acetylcholine**



Parasympathetic and cholinergic Sympathetic

In 1934 Henry Dale identified in Acetylcholine the substance released by **parasympathetic vagus nerve**.

- **Norepinephrine (Noradrenaline)**



Noradrenergic/Adrenergic Sympathetic

In the ~1930 Walter Cannon disclosed the **link between sympathetic nervous system and** adrenaline-expressing **adrenal medulla**

In 1946 Ulf von Euler identified in **Noradrenaline** the major neurotransmitter released by the sympathetic nerves (**Nobel prize in 1970**)

- **Epinephrine (Adrenaline)**

Different neurotransmitters are mediating different effects in the target organ

Differential responses of ANS transmitters

Examples:

• Acetylcholine receptors

Nicotinic **Excitatory**-It is a multi-subunit Cation (Na, K, Ca) channel that opens

Muscarinic M1, M3, M5 **Excitatory**-Opens calcium channels

Muscarinic M2, M4 **Inhibitory**-Mainly presynaptic

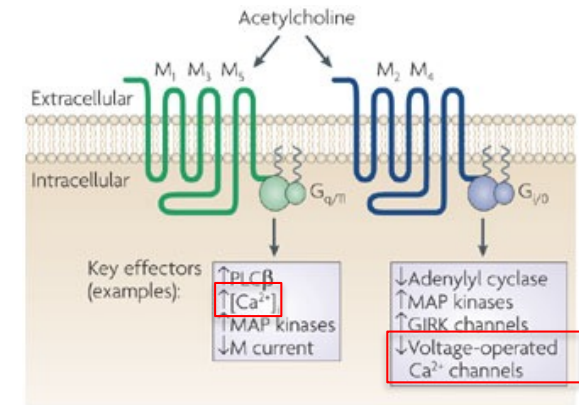
• Noradrenergic receptors

α_{1A-B-D} **Excitatory-Postsynaptic**—Phosphoinositol- Opens calcium channels

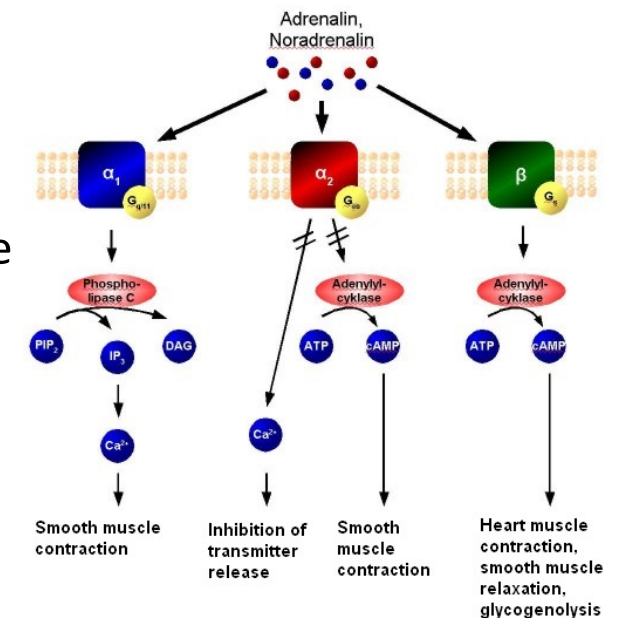
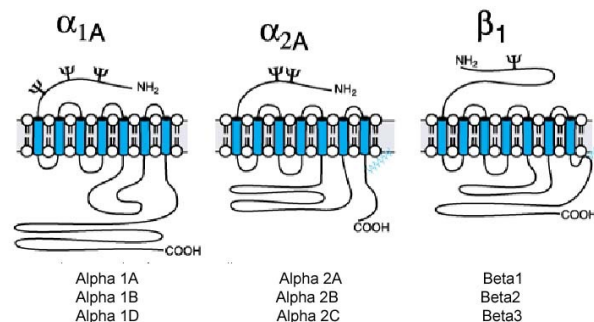
α_{2A-B-C} **Inhibitory**-Mainly *presynaptic*-G protein direct-

β_1 **Excitatory-Postsynaptic**-Adenylate cyclase-Opens calcium channels

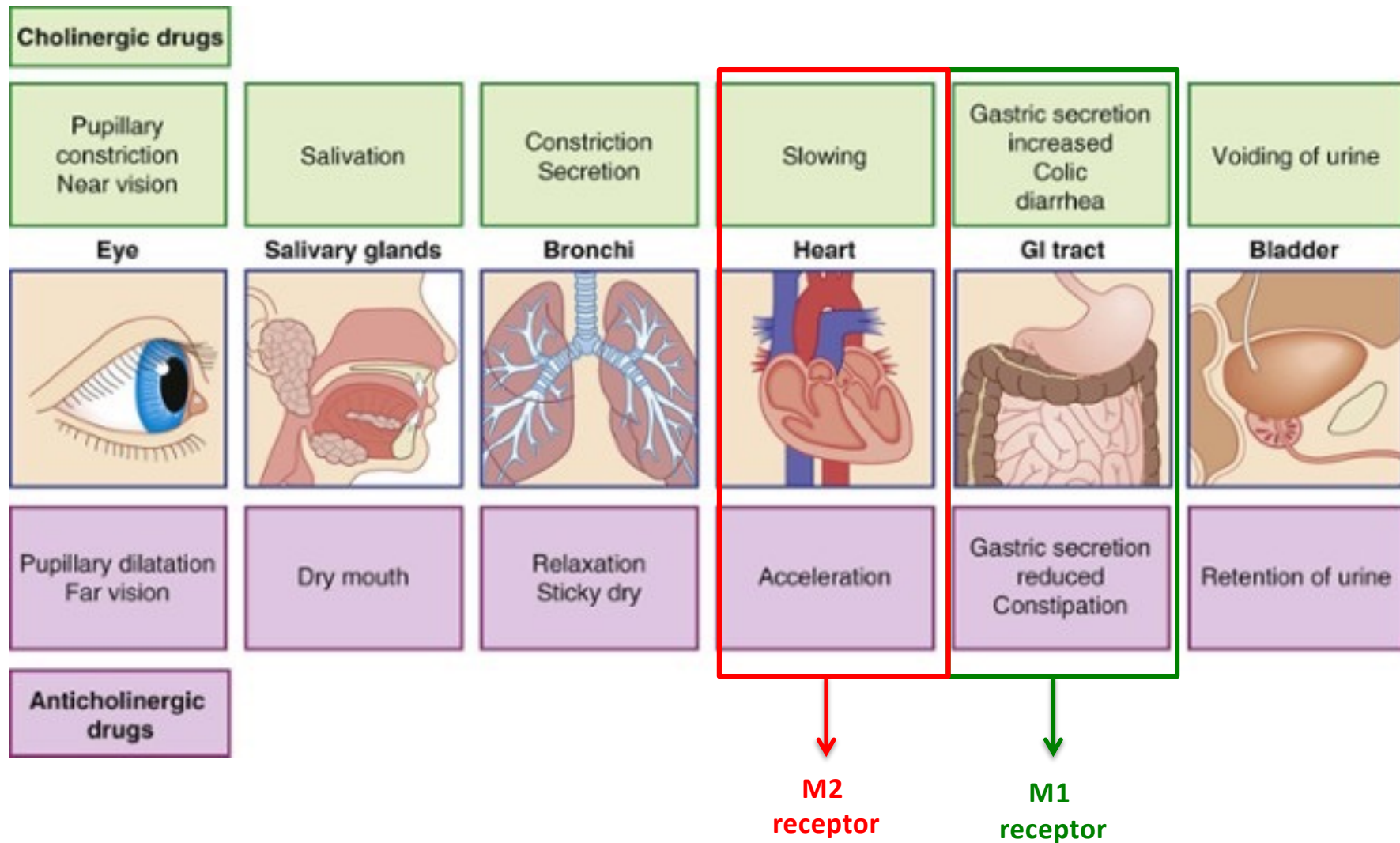
β_{2-3} **Inhibitory-Post- and pre-synaptic**--Adenylate cyclase



Nature Reviews | Drug Discovery



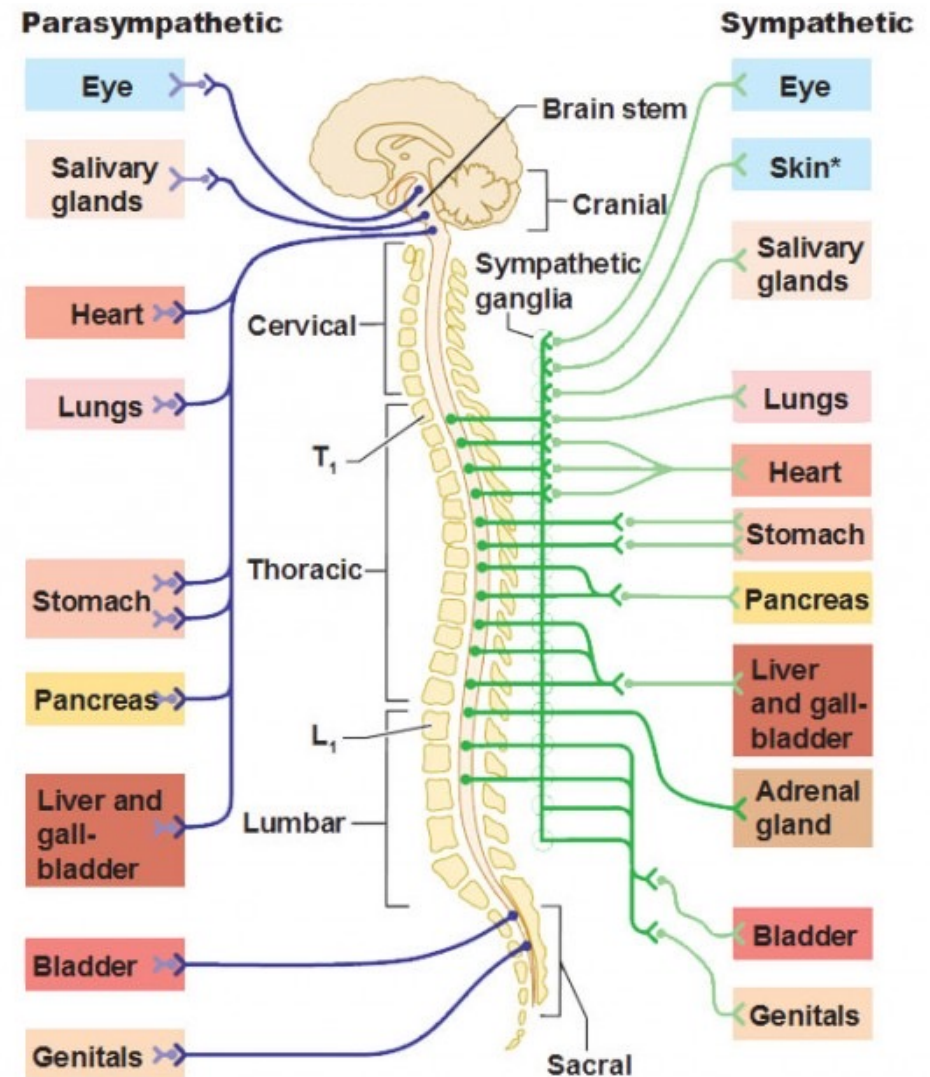
Tissue/organ-specific drugs



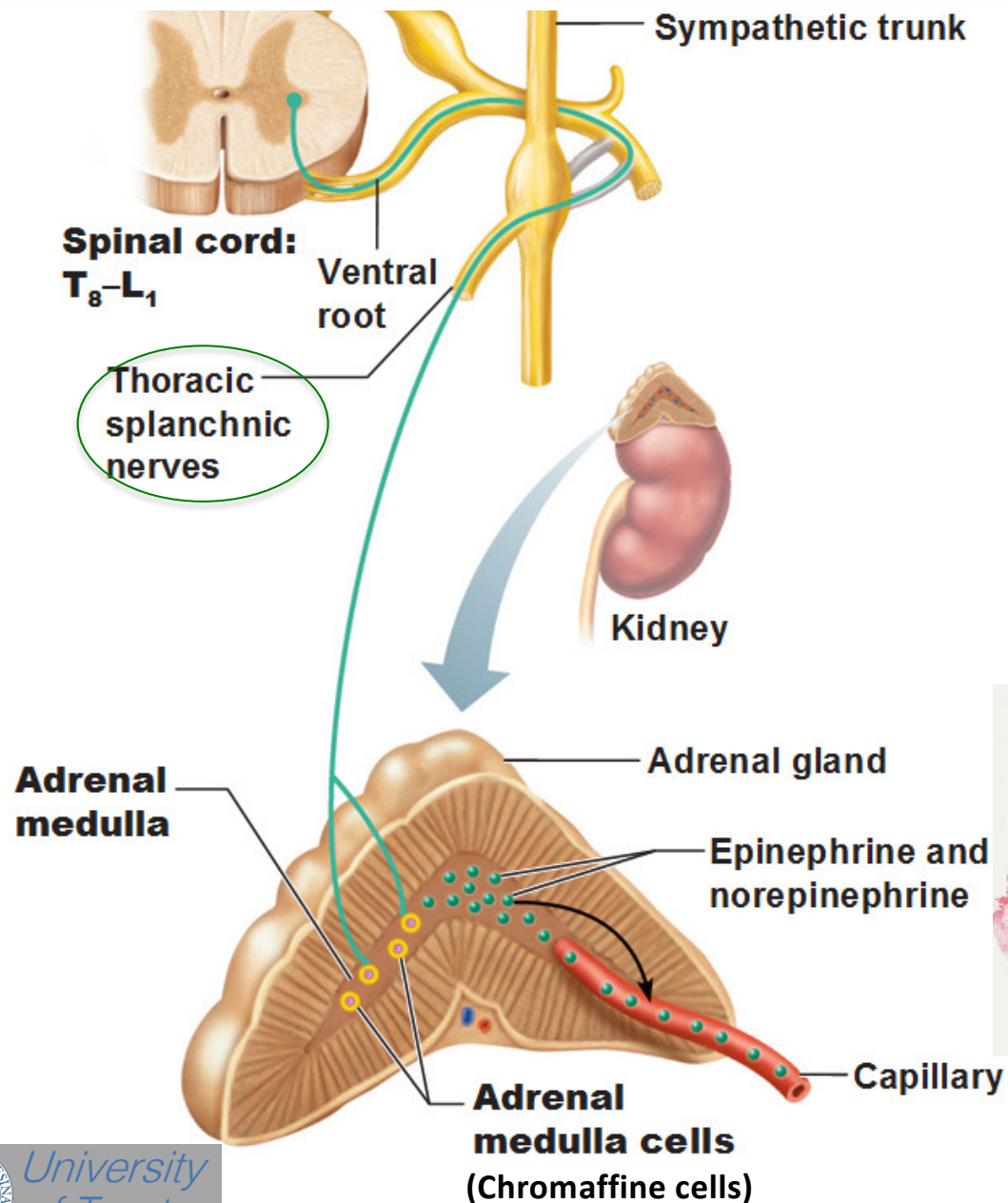
M1 specific blockers can be used to **reduce gastric acidity** without inducing tachycardia

Autonomic Nervous System and Nerves

- **Parasympathetic → Cranio-sacral System** → Cranial nerves:
 - III (oculomotor),
 - VII (facial),
 - IX (glossopharyngeal),
 - X (vagus)Sacral S3, S3, S4
- **Sympathetic → Thoraco-lombar system**
→ from Thoracic T1 to lumbar L2



Adrenal medulla and the sympathetic system

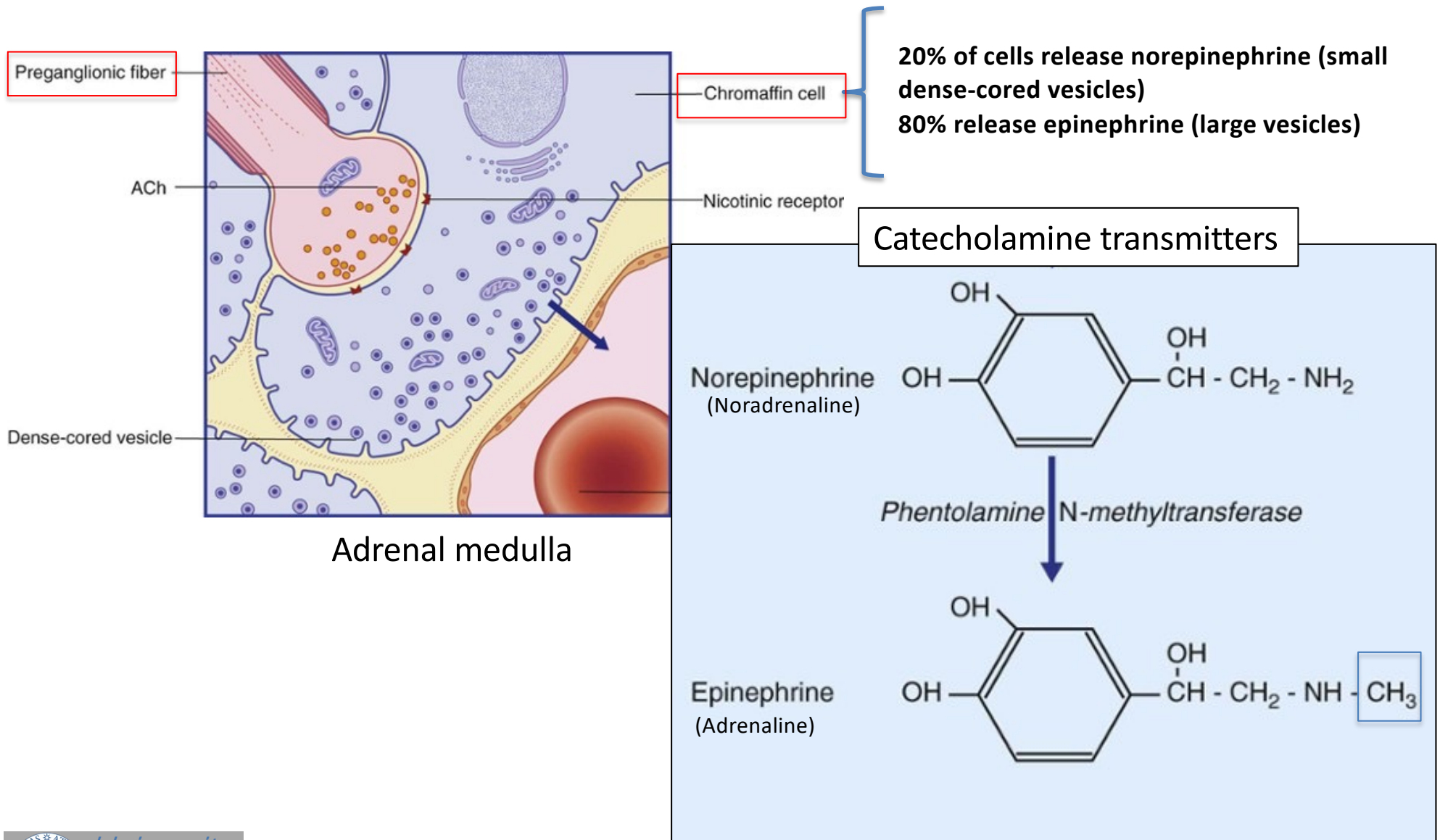


Chromaffine cells of the adrenal medulla:

- may be seen as **modified postganglionic sympathetic neurons**



Chromaffin cells release Norepinephrine and Epinephrine



The major role for ANS is the maintenance of Homeostasis

The different components of the autonomic nervous systems participate in maintaining the homeostasis of the organism

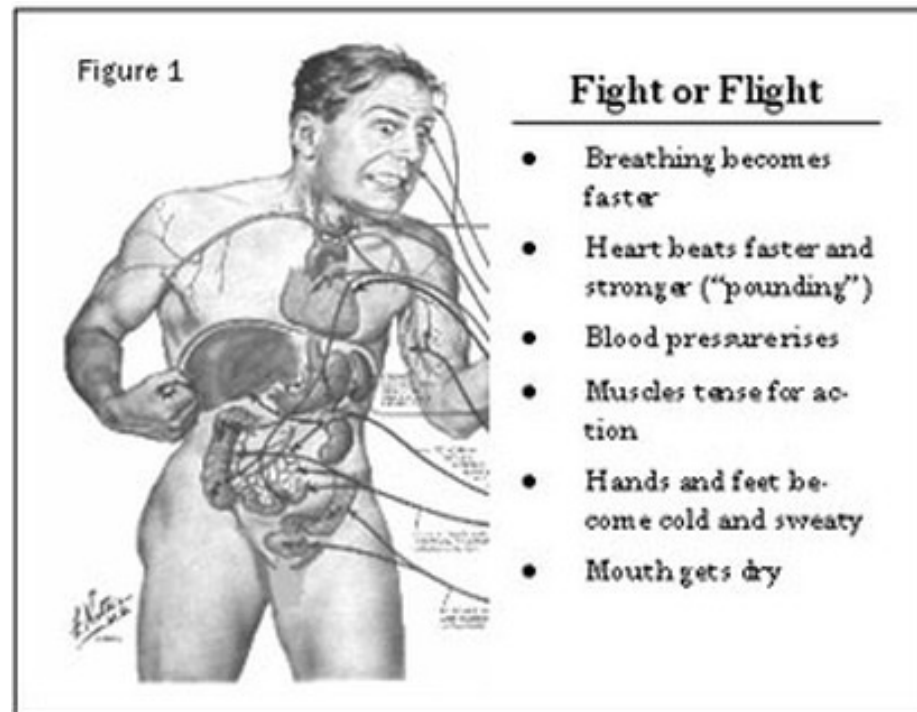
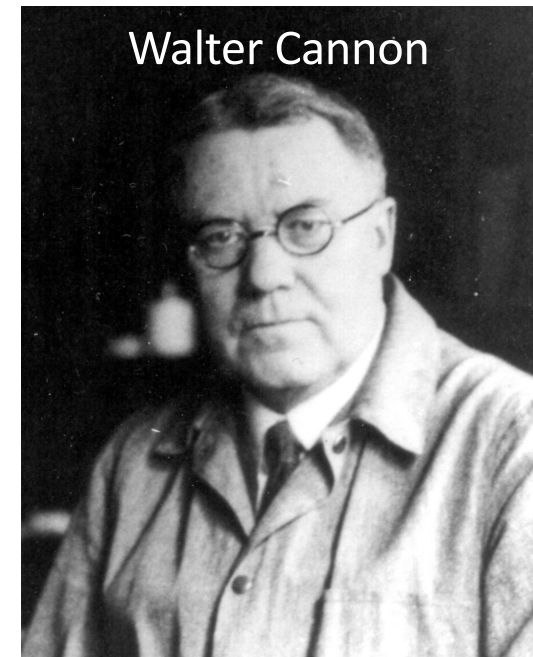


Homeostasis and fight-or-flight response

In 1865 Claude Bernard postulated that **the body maintains the constant internal environment by myriad, continual, compensatory reactions**. These compensatory reactions tend to restore a state of equilibrium in response to any outside changes, enabling independence from the external environment.

In 1929 Cannon extended the concept of Bernard and coined the word ***Homeostasis***

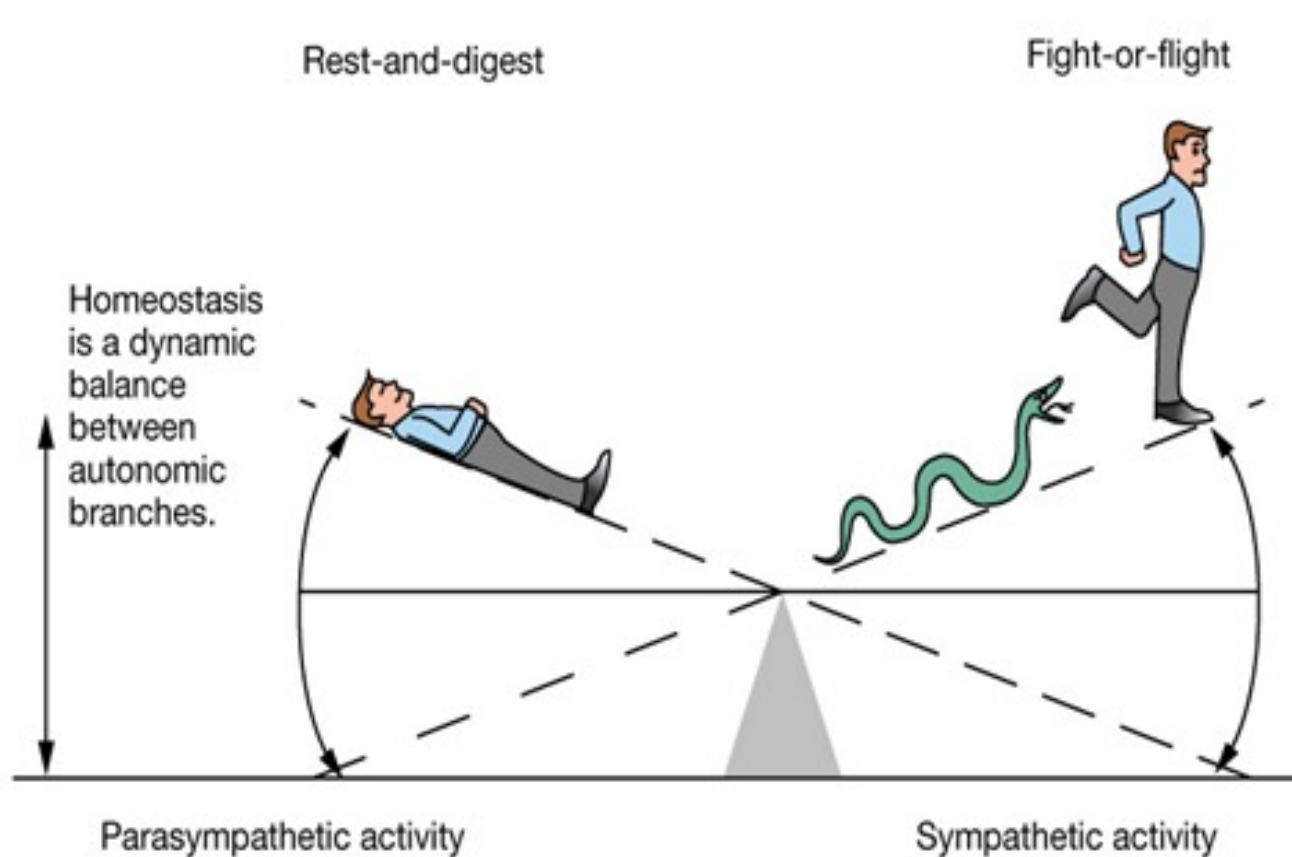
Cannon described also the **fight-or-flight response** and linked the autonomic nervous system to it



Fight-or-Flight response and Homeostasis

A different contribution from the different components is required when the homeostasis of the organism is challenged by external factors or functional demand

Also the fight-or-flight response may be seen as related to homeostasis



Major functions of the different components of the ANS

Sympathetic noradrenergic nervous system

- Is strongly induced during the **“Fight or flight” response** (increased heart rate, sweat, dilated pupils, cutaneous vasoconstriction, muscle vessel dilatation..)
- Also in absence of emergency **norepinephrine** is released from nerve terminals and **with rapid adjustment contribute to “normal” life functions**, such as walking...



Adrenal medulla

- Releases Adrenaline in the **bloodstream** (not organ specific as the sympathetic branch)
- Adrenaline is the first hormone to be identified and produced in a laboratory
- **Adrenaline levels change** not only in fight or flight response, but also in **normal situations**: i.e. adrenaline doubles just by **standing-up**

Parasympathetic nervous system

- Its activity is associated with the **“Rest and Digest” response**
- It **counterbalances the effect of (nor)adrenaline** in many districts



Different ANS components are finely controlled

- **Different stimuli elicit different responses**
 - **localized responses** (only one or few tissues)
 - **systemic responses**
 - Predominantly be dependent on one component (**sympathetic vs parasympathetic**)

For example:

- The **adrenergic system dominates the response to global or metabolic treat** such as glucoprivation and emotional stress, whereas **the noradrenergic sympathetic system dominates more localized responses** such as response to cold or to exercise
 - The response to **“fight”** is not exactly the same as in **“flight”**, and certainly different from **“rest”**
- **Sometime sympathetic and** parasympathetic nervous system **collaborate** to the same physiological function
 - **Different effects both necessary for organ functioning** (for example in the control of urine secretion cycle)
 - **Same effect** (for example both promote airways mucus secretion)

Summary

- The **Peripheral NS** is divided in different branches: **sensory, somatic motor, autonomic**
- PNS Axons (fibers) are organized in **nerves: cranial and spinal nerves**
- **Segmental distribution** of sensory/motor innervation (Dermomatome)
- **Nerve Plexuses** reorganize axons exiting spinal cord in different nerve terminal (redundancy)
- The **autonomic nervous system (ANS)** has two branches: **Parasympathetic and Sympathetic**
 - participates in maintaining the homeostasis
 - often opposite effects, but sometimes synergize
 - **SNS predominates in fight or flight**, whereas **parasympathetic NS** in **rest and digest**
- **Similarities: both** relay on **two neurons** to connect CNS and peripheral tissues (pre- and post-ganglionic neuron), in both the **pre-ganglionic neurotransmitter is Ach**
- **Major Differences:**
 - **Parasympathetic:**
 - originates from **cranio-sacral level**
 - **post-ganglionic** neurons are secreting **Ach**
 - **Sympathetic:**
 - originates from **thoraco-lumbar level**,
 - **post-ganglionic** neurons are generally secreting **NE**
 - **Pre-ganglionic Sympathetic neurons** synapse with **chromaffin cells of the adrenal medulla** to secrete **Adrenaline (80%)** and **NE (20%)** in the **blood stream**

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Chapters: 13

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Sai & Krarup E,; Elsevier

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Chapter: 2,3